TEAM AI_BYTE · CHIPATHON 2026

Layout Review

Reconfigurable AI & Mathematical Expression Accelerator for Edge Inference

Digital core dry-run on GF180MCU (gf180mcuD) · LibreLane 3.0

Status: WIP checkpoint for organizer layout review · August 2026

Design goal

- Lightweight **edge AI / math accelerator** for GF180MCU open PDK
- Host interface: 8-bit memory-mapped bus (addr / data / WE / RE)
- Compute: 4×4 systolic CNN tiles, Q8.8 ALU, EML (sigmoid, tanh, recip, sqrt, softmax)
- Single clock domain, 5 V core + I/O
- Full RTL → LibreLane → GDS path demonstrated on digital core

> This review covers the **core-only** hardened macro (`ai_byte_top`). Final submission will be workshop-slot `chip_top.gds` with padring.

PDK
gf180mcuD
wafer-space 1.8.0

STDCELL
mcu9t5v0
5 V library

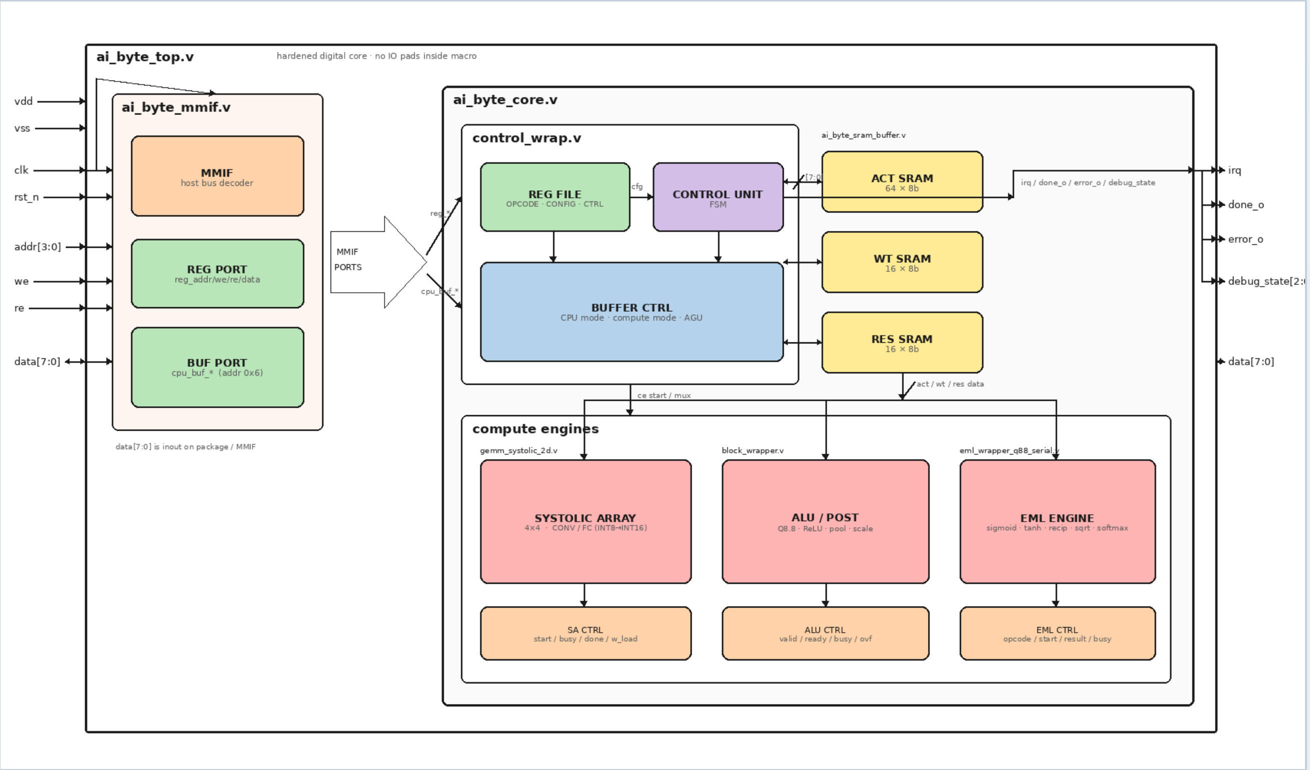
TARGET CLOCK
10 MHz
100 ns period

CORE DIE
1100^2 µm
$1.21\ mm^2$

Top-level wrapper

- Digital design hardened as a single macro: `ai_byte_top`
- Core config omits IO pads (bare MMIF ports)
- Pad adapter `chip_core.sv` + workshop padring → `chip_top`
- Flow: LibreLane 3.0 Classic · `config_ai_byte_core.yaml`
- Functional pins: CLK, RST_N, ADDR[3:0], DATA[7:0], WE, RE, IRQ, DONE, ERROR + VDD/VSS

CORE MACRO **DRY-RUN GDS**



ai_byte_top → MMIF + ai_byte_core (control, SRAMs, compute engines)

Initial Yosys synthesis

Metric	Value
Standard Cell PDK	gf180mcu_fd_sc_mcu9t5v0
Mapped standard cells	~19,020
Flip-flops	3,178
Synth chip area	~645,175 µm² (~0.645 mm²)
Sequential share of area	~40%
Synth check errors	0

From run `RUN_2026-08-06_13-06-45` · `make librelane-core`

Pre-PnR STA (period_min)

Corner	period_min	≈ fmax
nom_ss_125C_4v50	72.61 ns	13.8 MHz
nom_tt_025C_5v00	36.72 ns	27.2 MHz
nom_ff_n40C_5v50	7.88 ns	127 MHz

25 MHz (40 ns) does not close on SS. Target set to **10 MHz / 100 ns** for PnR margin; tighten after RTL pipelining.

Floorplan, timing & optimizer config

Parameter	Value
DIE_AREA	1100 × 1100 µm
CORE_AREA	10 … 1090 µm (10 µm margin)
PL_TARGET_DENSITY_PCT	55%
CLOCK_PORT / NET	clk
CLOCK_PERIOD	100 ns (10 MHz)
GRT_ALLOW_CONGESTION	true
Hold slack margins	PL 0.35 / GRT 0.3

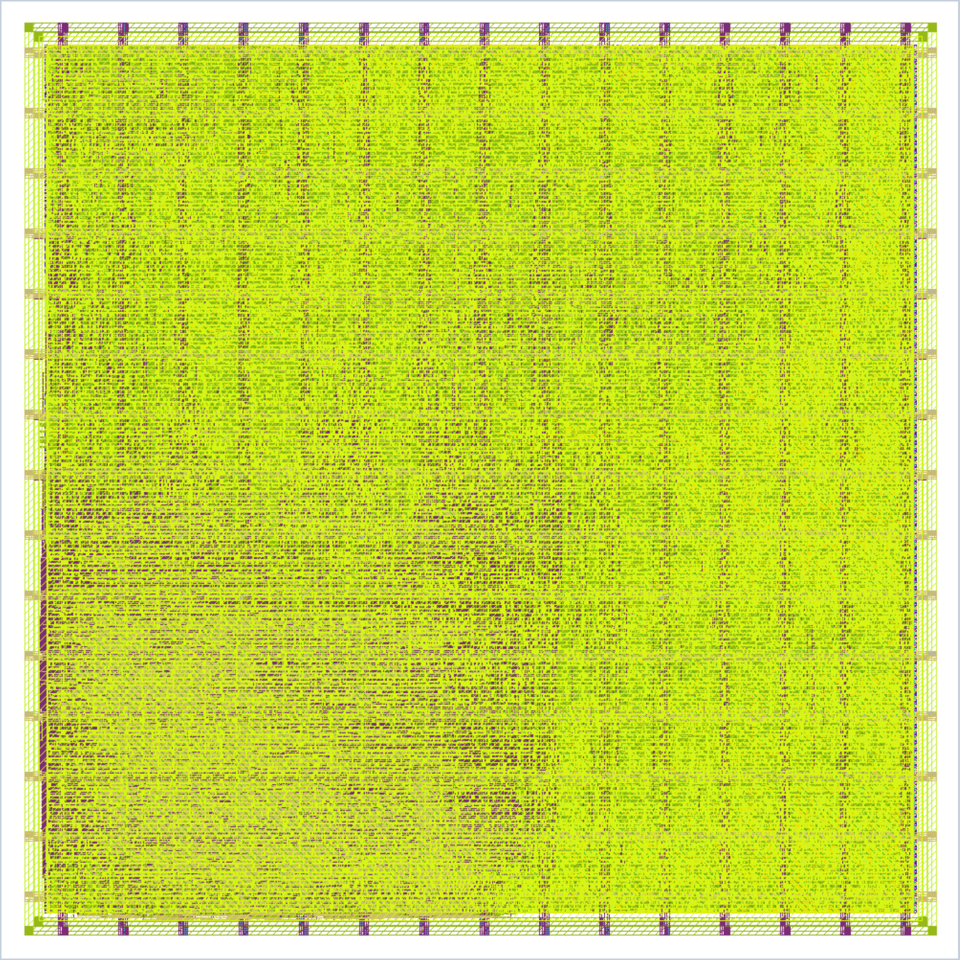
Routing / antenna (dry-run)	Value
DRT_OPT_ITERS	15 (default 64)
DRT_ANTENNA_REPAIR_ITERS	0
ERROR_ON_MAGIC_DRC	False

DRT iters and antenna repair were deliberately capped to produce a same-day dry-run GDS. This is **not** a manufacturable signoff layout.

Power distribution and routing

- PDN straps: width 5 µm, spacing 1 µm, pitch 75 µm
- Core ring on (10 µm); **not** connected to pads in core-only flow
- Vertical straps on Metal2, horizontal on Metal3
- Primary streamout: KLayout
- Power-grid violation count: **0**

Net	Role
VDD	5 V single supply
VSS	Ground



Core layout render (ai_byte_top) after LibreLane Classic flow

Final macro layout (core dry-run)

Metric	Value
Die / core area	1.21 / 1.165 mm^2
Instance count (all)	50,734
Stdcell count	29,915
Sequential cells	3,178
Stdcell utilization	~81.7%
Routed wirelength	~2.19e6 µm
Antenna diodes inserted	70
IO pins (core ports)	24

Signoff status (not clean):
Routing DRC 706 · KLayout DRC 134 · Magic DRC 146 · LVS errors 323 · XOR 3 · Antenna nets 17

Root cause: DRT stopped at iter 15 with shorts/spacing still open (15893 → 706). Downstream DRC/LVS noise follows from unfinished routing.

GDS produced and RTL→GDS path validated. Views under `final/` / `gds/ai_byte_top.gds`.

Static timing analysis (100 ns target)

Corner (setup WS)	Worst slack	Setup viol.
nom_tt_025C_5v00	+39.5 ns	0
nom_ff_n40C_5v50	+55.1 ns	0
nom_ss_125C_4v50	−7.7 ns	53
max_ss_125C_4v50 (global worst)	−19.8 ns	150

Hold: **positive** on all reported corners (0 hold violations).

- TT / FF close with healthy setup margin at 10 MHz
- SS still failing after incomplete/messy routing
- Global setup WNS ≈ −19.8 ns · TNS ≈ −451 ns
- Max slew / fanout / cap violations elevated (esp. SS) — expect improvement once DRT finishes cleanly

Next: clean routing first, then recover SS via density/die size and optional RTL piping / multi-cycle paths.

Voltage drop, power, and antenna

TOTAL POWER **58.4 mW** estimate from LibreLane	INTERNAL **41.3 mW** switching 17.1 · leakage ~7 µW
WORST IR DROP **0.185 mV** VDD → 4.99982 V @ nom_tt	ANTENNA **17 nets** 70 diodes; repair iters = 0

- IR drop is negligible at reported corner (<0.2 mV)
- Power dominated by internal + switching (leakage tiny)
- Antenna: post-DRT antenna repair disabled for dry-run turnaround
- Will re-enable limited `DRT_ANTENNA_REPAIR_ITERS` on signoff runs

> Power/IR numbers are from the dry-run layout and should be re-checked after a clean route.

Padring integration — pin mapping

Function	Direction	Pad type (planned)
VDD / VSS	power	dvdd / dvss pads
CLK	input	gf180mcu_fd_io__in_s (Schmitt)
RST_N	input	gf180mcu_fd_io__in_c
ADDR[3:0], WE, RE	input via bidir	bi_24t (OE=0)
DATA[7:0]	bidirectional	bi_24t
IRQ, DONE, ERROR	output via bidir	bi_24t (OE=1)

- Workshop slot: die 2935×2935 µm, core ~2051×2051 µm
- Mapping implemented in `chip_core.sv` (bidir[19:0])
- ADDR → bidir[3:0]; DATA → bidir[11:4]; WE/RE → bidir[13:12]; IRQ/DONE/ERROR → bidir[16:14]
- CLK / RST_N use dedicated input pads in `chip_top.sv`

> Functional minimum ≈ 19 digital + power/ground. Workshop slot provides many spare analog / bidir pads.

Padring integration — status

- Template: chipathon workshop padring (`SLOT=workshop`)
- Configs ready: `librelane/config.yaml` + `slots/slot_workshop.yaml`
- Command for full chip: `SLOT=workshop make librelane`
- Core dry-run used `make librelane-core` (this review)

> Full-chip padframe PnR + clean DRC/LVS is **in progress** — not yet claimed clean like a final tapeout package. Organizer dry-run currently points at core GDS.

Verification status

Item	Status
RTL cocotb (pad-level e2e)	In place / passing suite
Golden model co-sim	Yes (opcodes + tolerances)
Core LibreLane → GDS	Done (WIP signoff)
Workshop chip_top GDS	Next milestone
Gate-level + SDF sim	After clean PnR

What we will fix before final / Channel Partner

- Raise routing budget (`DRT_OPT_ITERS` toward 64)
- Re-enable limited antenna repair after DRT
- Ease congestion: larger die and/or lower placement density
- Recover SS setup once routes are clean (period / piping / multi-cycle)
- Integrate into workshop padframe → submit `chip_top.gds`
- Run GL + SDF post-layout simulation

NOW
Core dry-run
RTL→GDS proven

NEXT
Clean DRT
DRC / LVS / STA

THEN
Workshop pads
chip_top.gds

FINALLY
GL + SDF
signoff package

> Feedback from this review is welcome — especially on die size vs density trade-offs for GF180MCU congestion.

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Thank you!

Any feedback is appreciated.

Repo: chipathon-2026-AI_Byte

Artifacts: docs/dry-run-layout-report.md · librelane/runs/RUN_2026-08-06_13-06-45

Chipathon 2026 · GF180MCU